Akoizo Original Practice - Form 2 - Answer key

Division B | Circuit Lab | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

ANSWER KEY - Review only after submitting.

1. (2 points) An ampere is…

Unit: Current, voltage, resistance, power, and energy.
Worked-case data: A 6 V source drives a 300 Ω resistor for 60 s. Find current, power, and energy.

A. Joule per coulomb

B. Ohm per meter

C. Coulomb per second

D. Joule per second

Correct: C

2. (2 points) XOR(1,1) equals…

A. 2

B. Undefined

C. 0

D. 1

Correct: C

3. (2 points) Equal LED current guarantees equal brightness?

Unit: LED matching, charge-time estimates, and troubleshooting.
Worked-case data: Estimate ideal time for 1.5 Ah supplied at constant 0.30 A.

A. Only for any red LED

B. Only at 5 V

C. No

D. Always

Correct: C

4. (2 points) A watt-hour measures…

Unit: Current, voltage, resistance, power, and energy.
Worked-case data: A 6 V source drives a 300 Ω resistor for 60 s. Find current, power, and energy.

A. Resistance

B. Charge alone

C. Energy

D. Current

Correct: C

5. (2 points) NOT(A AND B) equals…

Unit: Digital logic and combinational circuits.
Worked-case data: For A=1, B=1, find half-adder sum and carry.

A. (NOT A) AND B

B. A AND B

C. (NOT A) OR (NOT B)

D. A OR B

Correct: C

6. (2 points) 1.5 Ah / 0.30 A equals…

A. 5 J

B. 0.20 h

C. 5 h

D. 0.45 h

Correct: C

7. (2 points) At fixed R, doubling V changes power by…

Unit: Current, voltage, resistance, power, and energy.
Worked-case data: A 6 V source drives a 300 Ω resistor for 60 s. Find current, power, and energy.

A. A factor of one half

B. No change

C. A factor of four

D. A factor of two

Correct: C

8. (2 points) A multiplexer…

Unit: Digital logic and combinational circuits.
Worked-case data: For A=1, B=1, find half-adder sum and carry.

A. Stores a clocked bit by definition

B. Measures resistance

C. Selects an input

D. Always adds inputs

Correct: C

9. (2 points) Useful troubleshooting changes…

Unit: LED matching, charge-time estimates, and troubleshooting.
Worked-case data: Estimate ideal time for 1.5 Ah supplied at constant 0.30 A.

A. Only the answer label

B. All components randomly

C. One suspected factor at a time

D. Every wire at once

Correct: C

10. (4 points) Why is charge not consumed?

Unit: Current, voltage, resistance, power, and energy.
Worked-case data: A 6 V source drives a 300 Ω resistor for 60 s. Find current, power, and energy.

answer: Steady charge flow conserves charge while energy changes form. (4 points).

11. (4 points) Give half-adder equations.

Unit: Digital logic and combinational circuits.
Worked-case data: For A=1, B=1, find half-adder sum and carry.

answer: Sum = A XOR B; carry = A AND B. (4 points).

12. (4 points) Why separate branch resistors?

Unit: LED matching, charge-time estimates, and troubleshooting.
Worked-case data: Estimate ideal time for 1.5 Ah supplied at constant 0.30 A.

answer: Different LED characteristics can produce unequal current in uncontrolled shared branches. (4 points).

13. (4 points) Convert 2 Wh to joules.

answer: 7200 J. (4 points).

14. (4 points) Why test all input combinations?

Unit: Digital logic and combinational circuits.
Worked-case data: For A=1, B=1, find half-adder sum and carry.

answer: Different cases can expose omitted inversions or incorrect branch logic. (4 points).

15. (4 points) Why can real charging exceed the ideal time?

Unit: LED matching, charge-time estimates, and troubleshooting.
Worked-case data: Estimate ideal time for 1.5 Ah supplied at constant 0.30 A.

answer: Current tapers and efficiency, temperature, and control constraints matter. (4 points).

16. (4 points) Why does divider loading matter?

Unit: Diagrams, nodes, series, parallel, and Ohm’s law.
Worked-case data: Find Req for 100 Ω and 300 Ω in parallel on 6 V.

answer: The new branch changes equivalent resistance and the voltage distribution. (4 points).

17. (4 points) Explain a nodal branch current.

Unit: Kirchhoff laws and network problem solving.
Worked-case data: At a node, 12 mA and 8 mA enter. One branch leaves. Find its current.

answer: Current from node a to b through R is (Va - Vb)/R. (4 points).

18. (2 points) Define “Open circuit” in the context of Circuit Lab.

answer: A broken conducting path. (2 points).